

1. Show that $m(a+bX) = a + b \cdot m(X)$

$$\begin{aligned} m(a+bX) &= \frac{1}{N} \sum_{i=1}^N (a + bx_i) \\ &= \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N x_i \right) \\ &= \frac{Na}{N} + b \frac{1}{N} \sum_{i=1}^N x_i \\ &= a + b \cdot m(X) \end{aligned}$$

$$\therefore m(a+bX) = a + b m(X)$$

Q1.2 Show that $\text{cov}(X, a+bY) = b \cdot \text{cov}(X, Y)$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) [(a + by_i) - m(a+bY)]$$

$$\therefore m(a+bY) = a + b \cdot m(Y)$$

$$\therefore (a + by_i) - m(a+bY) = (a + by_i) - (a + b m(Y)) = b(y_i - m(Y))$$

$$\therefore \text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b(y_i - m(Y))$$

$$= b \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (y_i - m(Y))$$

$$= b \text{cov}(X, Y)$$

$$\therefore \text{cov}(X, a+bY) = b \text{cov}(X, Y)$$

Q1.3 Show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$
and in particular that $\text{cov}(X, X) = S^2$

$$\therefore \text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N [(a+bX_i) - m(a+bX)]^2$$

From Q1.1 : $m(a+bX) = a + b \cdot m(X)$

$$\therefore (a+bX_i) - m(a+bX) = b(X_i - m(X))$$

$$\therefore (a+bX_i) - m(a+bX) = b(X_i - m(X))$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N b^2 (X_i - m(X))^2$$

$$= b^2 \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2$$

$$= b^2 \text{cov}(X, X)$$

$$\therefore \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$$

$$\therefore \text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2 = S^2$$

$$\therefore \text{cov}(X, X) = S^2$$

Q1.4 Consider a non-decreasing transformation $g(\cdot)$

Is it always true that $m(g(X)) = g(m(X))$?

No. For example. let $X = (1, 3)$

$$g(X) = X^3$$

Since $g(X) = X^3$ is non-decreasing for all real X , it satisfies the condition.

$$m(X) = \frac{1+3}{2} = 2$$

$$g(m(X)) = 2^3 = 8$$

$$\text{However, } m(g(X)) = \frac{g(1) + g(3)}{2} = \frac{1^3 + 3^3}{2} = 14$$

$$\therefore m(g(X)) \neq g(m(X))$$

$\therefore$ it is not always true